

AUSTIN TONSAY

U.S. Citizen | +1 (562) 213-6508 | Cerritos, CA | <https://www.linkedin.com/in/austintonsay>

TECHNICAL SUMMARY:

Aerospace Engineering student at California State Polytechnic University, Pomona, with hands-on experience working with aircraft and model rocket structures, CubeSat thermal systems, and embedded avionics. I am looking for an internship position to apply my CAD design, control systems, and data-driven validation experience to real-world development.

EDUCATION:

California State Polytechnic University, Pomona

Bachelor of Science, Aerospace Engineering (Emphasis: Astronautics) | GPA: 3.83

Expected Graduation: May 2029

RELEVANT COURSEWORK:

ARO1011L - Constructed a 41.5 in. wingspan remote-controlled aircraft with micro servo-controlled control surfaces in a 6-person team to gather experimental thrust, lift, drag data and computed average flight velocity flying upwind and downwind

ARO1021L - Conducted comparative analysis between OpenRocket simulations, analytical calculations, and experimental altimeter flight data for rocket systems; identified discrepancies and sources of error

EGR1000L - Designed, programmed, and assembled a bipedal biomimetic robotic system as Project Engineer; integrated Arduino IDE (C++) control of 4 micro servo-actuated joints enabling 3 degrees of motion (vertical, horizontal, rotational).

PROJECTS:

Personal Project

May 2026 - August 2026

- Designed, manufactured, and compared a reference rocket and altitude-optimized rocket powered by H-class motors on a 2-person team while obtaining L1 certifications for each member
- Analyzed and compared performance data of drag, mass, and apogee gathered from OpenRocket, ANSYS, and SolidWorks simulations to real-world flight to determine which geometrical change influenced greatest apogee increase
- Assigned design freeze dates, milestone deadlines, and overall project deadlines to account for delivery and fabrication times

EXPERIENCE:

Design, Build, Fly (DBF) Competition:

August 2025 - Present

Structures/Manufacturing Member

- Developed 3 structured trade studies comparing 4-6 candidate materials each to optimize structural strength, weight, cost, and manufacturability for aircraft wing and fuselage components.
- Designed tail assembly in SolidWorks intended for 3D printing to minimize part count, enhance manufacturability, and improve aerodynamic performance
- Fabricated wing/fuselage structure using SolidWorks-driven layouts, wood/foam core construction with carbon-fiber reinforcement, and controlled surface finishing with Monokote heat-shrink covering.

Bronco Situation, Task, Action, Result (Bronco S.T.A.R.):

August 2025 - May 2026

Balloon Launch Assessment Directive for Everyone (BLADE) Odyssey, Communications & Thermals Member

- Developed a heating-pad trade matrix to evaluate 3 heater options against a $\sim 9.25\text{W} / 37.185\text{Wh}$ system power constraint for a 1U CubeSat gathering thermal coating data in near-space conditions at $\sim 100,000\text{ft}$.
- Designed and constructed a thermostat control loop; sized and integrated two $60 \times 40 \text{ mm}^2$ heating pads at $4.2\text{V} @ 3\text{W} (0.71\text{A})$ each within calculated power budget.

Undergraduate Missiles, Ballistics, and Rocketry Association (UMBRA):

September 2025 - November 2025

Small Rocket Competition, Team Lead

- Designed model rocket's fins/body dimensions in OpenRocket, SolidWorks CAD, and Bambu Studio; improved projected apogee from 2100ft to 2900ft ($\sim 38\%$ increase) through aerodynamic optimization.
- Led a 5-person team; coordinated 1-2 meetings per week, assigned member roles, stayed within 2 days ahead of milestones.
- Programmed Arduino IDE for avionics systems to read rocket altitude, surrounding pressure, and temperature accurately.

SKILLS:

SolidWorks, Computer-Aided Design (CAD), OpenRocket, 3D Printing, Arduino IDE, Bambu Studio, Modular Assembly, Composite Laminate Manufacturing, Data Analysis, Trade Studies, Project Management, Additive Manufacturing, Microsoft Office, Excel VBA

MEMBERSHIPS & AFFILIATIONS

American Institute of Aeronautics and Astronautics (AIAA)

September 2025 - Present

Bronco Situation, Task, Action, Result (Bronco S.T.A.R.)

August 2025 - May 2026

Maximizing Engineering Potential (MEP)

August 2025 - May 2026

Students for the Exploration and Development of Space (SEDS-USA)

September 2025 - December 2025